LABORATORY 3

EEE301: Signals and Systems

Title: Fourier Series and Fourier Transform

Section 1: Fourier series

Jean Baptiste Joseph Fourier had introduced the idea that 'any' periodic signals could be represented by harmonically related sinusoids (series of sines and cosines).

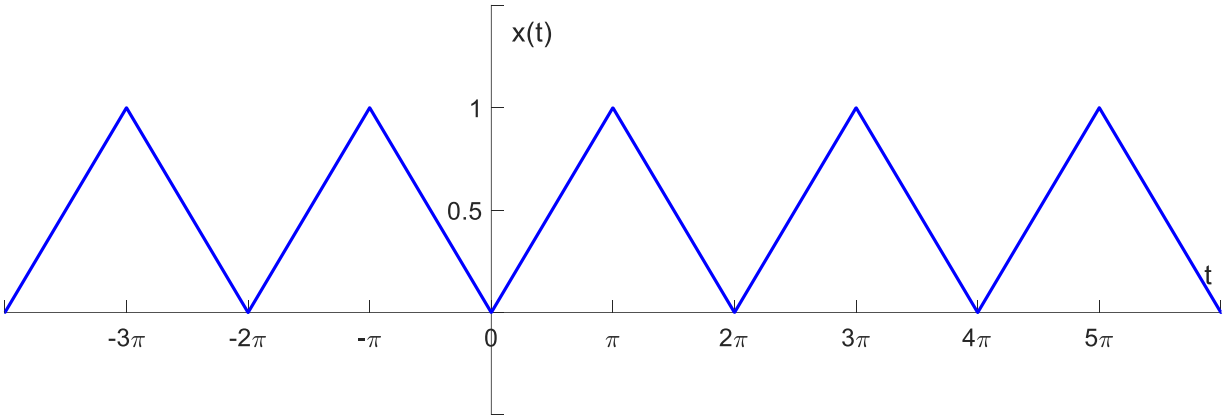


Figure 1: Triangular signal $x(t)$.

```
clc; clear; close all;

T = 8*pi;
fs = 100;
t = -T-1/fs:1/fs:T-1/fs;

x = 0.5 + 0.5*sawtooth(t,1/2);
plot(t,x,'LineWidth',2,'Color','b');
ylabel('x(t)'); xlabel('t');
ylim([-.5 1.5]); xlim([-4*pi 6*pi]);
box off;
ax = gca();
set(ax,'FontSize',16,'XAxisLocation','origin','YAxisLocation','origin','XTi
ck',(-4*pi:pi:6*pi),'XTickLabel',{'-4\pi','-3\pi','-2\pi','-
\pi','0','\pi','2\pi',...
    '3\pi','4\pi','5\pi','6\pi'});
```

Figure 2: MATLAB code for triangular signal $x(t)$

Figure 1 shows the a repetitive triangular signals with fundamental period T_0 and angular frequency ω_0. Given that,

$$x(t) = \frac{|t|}{\pi}, \quad \text{for } -\pi < t < \pi.$$

a) Find the fundamental frequency, ω_0 of the signals.

calculation

b) Fourier series of a periodic function is given by

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)\,; \quad t_1 < t < t_1 + T_0 \qquad (1)$$

The Fourier coefficients can be found from the following set of formula:

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt \qquad (2)$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos(n\omega_0 t)\, dt \qquad (3)$$

and

$$b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin(n\omega_0 t)\, dt \qquad (4)$$

Calculate the Fourier coefficients a_0, a_n and b_n for the signal $x(t)$.

calculation

c) Write the MATLAB code to plot the Fourier series, $x(t)$ that you get in (c) up until $n = 5$ terms.

```
Codes, 1 original x(t) plot and 5 Fourier series plots
```

d) Discuss your results obtained in Section 1(d)

```
Discussion
```

Section 2: Fourier Transform

Fourier analysis can be extended to non-periodic continuous time signals $x(t)$, using Fourier Transform (FT). Let consider a non-periodic rectangular signal

$$x(t) = \begin{cases} 1 & |t| \leq 0.5 \\ 0 & |t| > 0.5 \end{cases}$$

The signal $x(t)$ can be simulated as follows

```
N = 100;          % Sample number
Ts = 4/N;         % Sample for 4 seconds
fs = 1/Ts;        % Sampling rate
df = fs/N ;       % Set frequency-domain resolution
n = [-N/2:N/2-1]' ; % Vector of 2*N time indices
t = Ts*n ;        % Vector of times

width = 1;        % Rectangular width
x = rectpuls(t,width);  % Simulating rectangular function
plot(t,x,'LineWidth',2) ; grid on ;
xlabel('Time, t (s)') ; ylabel('x(t)') ;
```

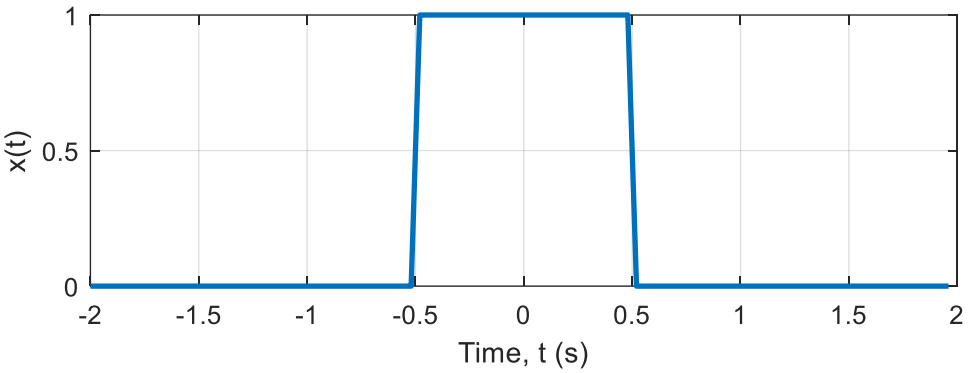


Figure 3: Rectangular signal $x(t)$

Continuous time Fourier transform of signal $x(t)$ can be approximated using '`fft`' MATLAB function as follows

```
k = [0:N-1]' ; % Vector of N frequency indices
X = Ts*fft(x) ; % Vector of 2N approx. X(f) CTFT
```

Plot of magnitude and phase plot of $X(f)$ can be obtain as

```
subplot(2,1,1);
plot(k*df,abs(X),'LineWidth',2) ; grid on ;
xlabel('Frequency Hz') ; ylabel('Magnitude, |X(f)|') ;

subplot(2,1,2) ;
plot(k*df,angle(X),'LineWidth',2) ; grid on ;
xlabel('Frequency Hz') ; ylabel('Phase, \angle X(f)') ;
```

The results of Magnitude and Phase plot is shown in Figure 4.

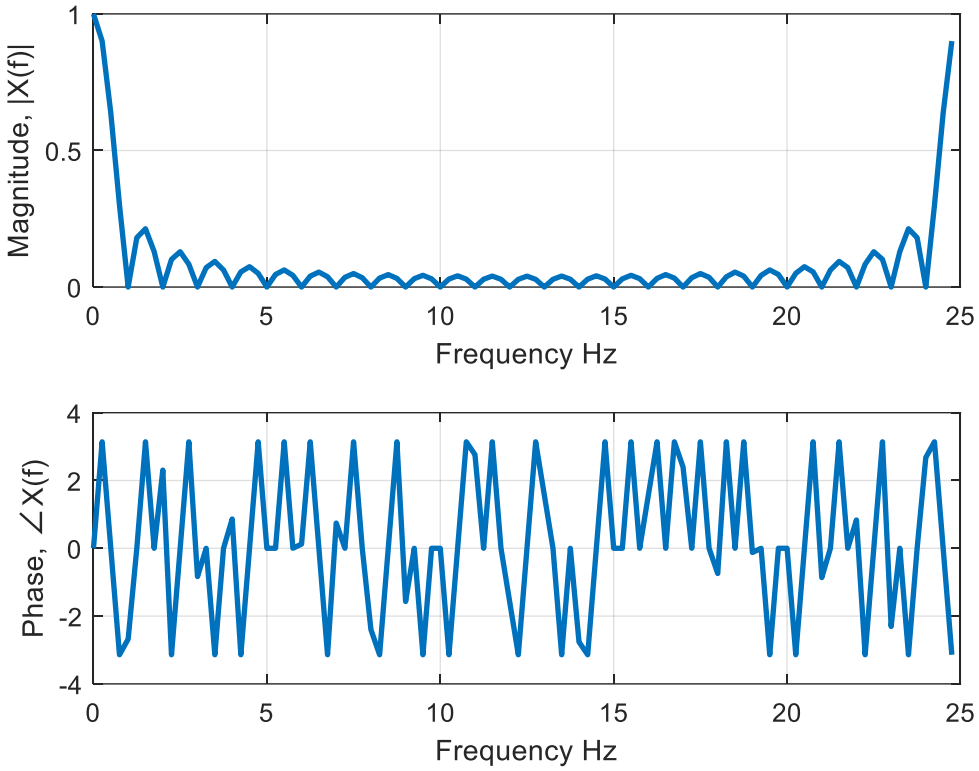


Figure 4: Magnitude and Phase plot for CTFT ($x(t) \rightarrow X(f)$).

Notice that N=100 samples are taken from the time-domain signal produce frequency domain [0 to 25Hz]. By using MATLAB function, `'fftshift'`,

a) Write your MATLAB code to obtain magnitude and phase plot of $X(f)$ with limit [-12.5Hz to 12.5Hz].

```
Codes and plots
```